USER INFORMATION
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FLUCONOL®
(Fluconazole 2 mg/mL Intravenous Infusion)

Product Description:
FLUCONOL® is a sterile intravenous solution containing Fluconazole 2mg/mL in iso-osmotic saline solution.
The solution is clear and colourless.

Composition:
Each mL solution contains:-
Fluconazole 2 mg in iso-osmotic saline solution.

Pharmacodynamic:
Fluconazole, a member of the triazole class of antifungal agents, is a potent and selective inhibitor of fungal enzymes necessary for the synthesis of ergosterol. Fluconazole shows little pharmacological activity in a wide range of animal studies. Some prolongation of pentobarbitone sleeping times in mice (p.o), increased mean arterial and left ventricular blood pressure and increased heart rate in anaesthetised cats (i.v) occurred. Inhibition of rat ovarian aromatase was observed at high concentrations.
Intravenously administered fluconazole was active in a variety of animal fungal infection models. Activity has been demonstrated against opportunistic mycoses, such as infections with Candida spp. including systemic candidiasis in immunocompromised animals; with Cryptococcus noeformans, including intracranial infections; with Microsporum spp. and with Trichophyton spp. Fluconazole has also been shown to be active in animal models of endemic mycoses, including infections with Blastomyces dermatitides; with Coccidoides immitis, including intracranial infection and with Histoplasma capsulatum in normal and immunosuppressed animals. There have been reports of cases of superinfection with Candida species other than C. albicans, which are often inherently not susceptible to fluconazole (e.g Candida krusei). Such cases may require alternative antifungal therapy.
Fluconazole is highly specific for fungal cytochrome P-450 dependent enzymes. Fluconazole 50mg daily given up to 28 days has been shown not to affect testosterone plasma concentrations in males or steroid concentrations in females of child-bearing age. Fluconazole 200-400mg daily has no clinically significant effect on endogenous steroid levels or on ACTH stimulated response in healthy male volunteers. Interaction studies with antipyrine indicate that single or multiple doses of fluconazole 50mg do not affect its metabolism.
The efficacy of fluconazole in tinea capitis has been studied in 2 randomised controlled trials in a total of 878 patients comparing fluconazole with griseofulvin. Fluconazole at 6 mg/kg/day for 6 weeks was not superior to griseofulvin administered at 11 mg/kg/day for 6 weeks. The overall success rate at week 6 was low (fluconazole 6 weeks: 18.3%; fluconazole 3 weeks: 14.7%; griseofulvin: 17.7%) across all treatment groups. These findings are not inconsistent with the natural history of tinea capitis without therapy.

Pharmacokinetics:
The pharmacokinetic properties of fluconazole are similar following administration by the intravenous or oral route. Administration of a loading dose (on day 1) of twice the usual daily dose enables plasma levels to approximate to 90% steady-state levels by day 2. The apparent volume of distribution approximates to total body water. Plasma protein binding is low (11-12%). Fluconazole achieves good penetration in all body fluids studied. The levels of fluconazole in saliva and sputum are similar to plasma levels. In patients with fungal meningitis, fluconazole levels in the CSF are approximately 80% of the corresponding plasma levels. High skin concentrations of fluconazole, above serum concentrations, are achieved in the stratum corneum, epidermis-dermis and eccrine sweat. Fluconazole accumulates in the stratum corneum. At a dose of 50mg once daily, the concentration of fluconazole after 12 days was 73 microgram/g and 7 days after cessation of treatment the concentration was still 5.8 microgram/g. The major route of excretion is renal, with approximately 80% of the administered dose appearing in the urine as unchanged drug. Fluconazole clearance is proportional to creatinine clearance. There is no evidence of circulating metabolites. The long plasma elimination half-life provides the basis for single dose therapy for genital candidiasis and once daily dosing for other indications.

Indications:
Therapy may be instituted before the results of the caltures and other laboratory studies are known; however, once these results become available, anti-infective therapy should be adjusted accordingly.
1. Cryptococcosis, including cryptococcal meningitis and infestions of other sides (e.g., pulmonary, cutaneous). Normal hosts and patients with AIDS, organ transplants or other causes of immunosuppression maybe treated. Fluconazole can be used as maintenance therapy to prevent relapse of cryptococcal disease in patients with AIDS.
2. Systemic candidiasis, including candidemia, disseminated candidiasis and other forms of invasive candidal infections. These include infections of the peritoneum, endocardium, eye, and pulmonary and urinary tracts. Patients with malignancy, in intensive care units, receiving cytotoxic or immunosuppressive therapy, or with other factors predisposing to candidal infection may be treated.
3. Mucosal candidiasis. These include oropharyngeal, esophageal, non-invasive broncopulmonary infections, candiduria, mucocutaneous and chronic oral atropic candidiasis (denture sore mouth). Normal hosts and patients with compromised immune function maybe treated. Prevention of relapse of oropharyngeal candidiasis in patients with AIDS.
4. Genital candidiasis. Vaginal candidiasis acute or recurrent; and propylaxis to reduce the incidence of recurrent vaginal candidiasis (3 or more episodes a year). Candidal balanitis.
5. Prevention of fungal infections in patients with malignancy who are predisposed to such infections as a result of cytotoxic chemotherapy or radiotherapy.
6. Dermatomycosis including tinea pedis, tinea corpuris, tinea cruris, tinea versicolor, and dermal candida infections.

Recommended Dosage:
The daily dose of Fluconazole should be based on the nature and severity of the fungal infection. Most cases of vaginal candidiasis respond to single dose therapy. Therapy for those types of infections requiring multiple dose treatment should be continued until clinical parameters or laboratory tests indicate that active fungal infection has subsided. An inadequate period of treatment may lead to recurrence of active infection. Patients with AIDS and cryptococcal meningitis of recurrent oropharyngeal candidiasis usually require maintenance therapy to prevent relapse.
Use in adults
1. For cryptococcal meningitis and cryptococcal infections at other sites, the usual dose is 400 mg on the first day followed by 200 mg-400 mg once daily. Duration of treatment for cryptococcal infections will depend on the clinical and mycological response, but is usually at least 6-8 weeks for cryptococcal meningitis. For the prevention of relapse of cryptococcal meningitis in patients with AIDS, after the patient receives a full course of primary therapy, fluconazole maybe administered indefinitely at daily dose of 200 mg.
2. For candidemia, disseminated candidiasis and other invasive candidal infections, the usual dose is 400 mg on the first day followed by 200 mg daily. Depending on the clinical response, the dose may be increased to 400 mg daily. Duration of treatment is based upon the clinical response.
3. For oropharyngeal candidasis, the usual dose is 50-100 mg once daily for 7-14 days. If necessary, treatment can be continued for longer periods in patients with compromised immune function. For atropic oral candidasis associated with dentures, the usual dose is 50 mg once daily for 14 days administered concurrently with local antiseptic measures to the denture. For other candidal infections of mucosa except genital candidasis [see below] (e.g., esophagitis, non-invasive broncopulmonary infections, candiduria, mucocutaneous candidiasis, etc.) the usual effective dose is 50-100 mg daily, given for 14-30 days. For the prevention of relapse of oropharyngeal candidiasis in patients with AIDS, after the patient receives a full course of primary therapy, fluconazole maybe administered at a 150 mg once weekly dose.
4. For the treatment of vaginal candidasis, fluconazole 150 mg should be administered a single oral dose. To reduce the incidence of recurrent vaginal candidiasis, a 150 mg once monthly dose maybe used. The duration of therapy should be individualized, but ranges from 4-12 months. Some patients may require more frequent dosing. For candida balanitis, fluconazole 150 mg should be administered as a single oral dose.
5. The recommended fluconazole dosage for the prevention of candidiasis is 50-400 mg once daily, based on the patient's risk for developing fungal infections. For patients at high risk of systemic infection e.g., patients who are anticipated to have profound or prolonged neutropenia, the recommended daily dose is 400 mg once daily. Fluconazole administration should start several days before the anticipated onset of neutropenia and continue for 7 days after the neutrophil count rises above 1000 cells per mm^3.
6. For dermal infections including tinea pedis, corpodis, cruris and candida infections the recommended dosage is 150 mg once weekly or 50 mg once daily. Duration of treatment is normally 2 to 4 weeks but tinea pedis may require treatment for up to 6 weeks. For tinea versicolor, the recommended dosage is 300mg once weekly for 2 weeks; a third weekly dose of 300 mg maybe needed in some patients, whereas, in some patients, a single dose of 300-400mg maybe sufficient. An alternate dosing regiment is 50 mg once daily for 2-4 weeks.

Use in Children
As with similar infections in adults, the duration of treatment is based on the clinical and mycological response. The maximum adult daily dosage should not be exceeded in children. Fluconazole is administered as a single dose each day. The recommended dosage of fluconazole for mucosal candidiasis is 3 mg/kg daily. A loading dose of 6 mg/kg may be used on the first day to achieve steady state levels more rapidly.
For the treatment of systemic candidiasis and cryptococcal infections, the recommended dosage is 6-12 mg/kg daily, depending on the severity of the disease.
For the prevention of fungal infections in immunocompromised patients considered at risk as a consequence of neutropenia following cytotoxic chemotherapy or radiotherapy, the dose should be 3-12 mg/kg daily, depending on the extent and duration of the induced neutropenia (see adult dosing).
For children with impaired renal function the daily dose should be reduced in accordance with the guidelines given for adults, dependent on the degree of renal impairment.
Children below 4 weeks of age
Neonates excrete fluconazole slowly. In the first two weeks of life, the same mg/kg dosing as in older children should be used but administered every 72 hours. During weeks 2-4 of life, the same dose should be given every 48 hours.
Use in elderly
Where there is no evidence of renal impairment, normal dosage recommendations should be adopted. For patients with renal impairment (creatinine clearance <50 ml/min) the dosage schedule should be adjusted as described below.
In patients with renal impairment
Fluconazole is predominantly excreted in the urine as unchanged drug. No adjustments in single-dose therapy are necessary. In patients with impaired renal function who will receive multiple doses of Fluconazole, an initial loading dose of 50 mg to 400 mg should be given. After the loading dose, the daily dose (according to indication) should be based on the following table:

Creatinine clearance (mL/min)	Percent of recommended dose
> 50	100%
11 - 50	50%
Patients receiving regular dialysis	100% after every dialysis session

Route Of Administration : Intravenous (IV)

Contraindication
FLUCONOL® should not be used in patients with known hypersensitivity to fluconazole or to related azole compounds or any other ingredient in the formulation. Fluconazole should not be co-administered with cisapride or terfenadine which are known to both prolong the QT-interval and are metabolised by CYP3A4.

Warning and Precautions
In some patients, particularly those with serious underlying diseases such as AIDS and cancer, abnormalities in haematological, hepatic, renal and other biochemical function test results have been observed during treatment with FLUCONOL® but the clinical significance and relationship to treatment is uncertain.
Very rarely, patients who died with severe underlying disease and who had received multiple doses of FLUCONOL® had post-mortem findings which included hepatic necrosis. These patients were receiving multiple concomitant medications, some known to be potentially hepatotoxic, and / or had underlying diseases which could have cuased the hepatic necrosis.
In cases of hepatotoxicity, no obvious relationship to total daily dose of FLUCONOL®, duration of therapy, sex or age of the patient has been observed; the abnormalities have usually been reversible on discontinuation of FLUCONOL® therapy. As a causal relationship with FLUCONOL® cannot be excluded, patients who develop abnormal liver function tests during FLUCONOL® therapy should be monitored for the development of more serious hepatic injury. FLUCONOL® should be discontinued if clinical signs or symptoms consistent with liver disease develop during treatment with FLUCONOL®. Patients have rarely developed exfoliative cutaneous reactions, such as Stevens-Johnson Syndrome and toxic epidermal necrolysis, during treatment with Fluconazole. AIDS patients are more prone to the development of severe cutaneous reactions to many drugs. If a rash develops in a patient treated for a superficial fungal infection which is considered attributable to FLUCONOL®, further therapy with this agent should be discontinued. If patients with invasive / systemic fungal infections develop rashes, they should be monitored closely and FLUCONOL® discontinued if bullous lesions or erythema multiforme develop.
In rare cases, as with other azoles, anaphylaxis has been reported. Some azoles, including fluconazole, have been associated with prolongation of the QT interval on the electrocardiogram. During post-marketing surveillance, there have been very rare cases of QT prolongation and torsade de pointes in patients taking fluconazole. Although the association of fluconazole and QTprolongation has not been fully established, fluconazole should be used with caution in patients with potentially proarrythmic conditions such as :
· Congenital or documented acquired QT prolongation.
· Cardiomyopathy, in particular when heart failure is present.
· Sinus bradycardia.
· Existing symptomatic arrythmias.
· Concomitant medication metabolised by CYP3A4 know to prolong QT interval.
· Electrolyte disturbances such as hypokalaemia, hypomagnesaemia and hypocalaemia.

Interactions With Other Medicaments
The following drug interactions relate to the use of multiple-dose fluconazole, and the relevance to single-dose 150mg fluconazole has not yet been established.
Rifampicin Concomitant administration of fluconazole and rifampicin resulted in a 25% decrease in the AUC and 20% shorter half-life of fluconazole. In patients receiving concomitant rifampicin, an increase in the fluconazole dose should be considered.
Hydrochlorothiazide In a pharmacokinetic interaction study, co-administration of multiple-dose hydrochlorothiazide to healthy volunteers receiving fluconazole increased plasma concentrations of fluconazole by 40%. An effect of this magnitude should not necessitate a change in the fluconazole dose regimen in subjects receiving concomitant diuretics.
Anticoagulants In an interaction study, fluconazole increased the prothrombin time (12%) after warfarin administration in healthy males. In post-marketing experience, as with other azole antifungals, bleeding events (bruising, epistaxis, gastrointestinal bleeding, hematuria and melaena) have been reported in association with increases in prothrombin time in patients receiving fluconazole concurrently with warfarin. Prothrombin time in patients receiving coumarin-type anticoagulants should be carefully monitored.
Benzodiazepines
Following oral administration of midazolam, fluconazole resulted in substantial increases in midazolam concentrations and psychomotor effects. This effect on midazolam appears to be more pronounced following with fluconazole administered intravenously. If concomitant benzodiazepine therapy is necessary in patients being treated with fluconazole, consideration should be given to decreasing the benzodiazepine dosage and the patients should be appropriately monitored.
Sulphonylureas
Fluconazole has been shown to prolong the serum half-life of concomitantly administered oral sulphonylureas (chlorpropamide, glibenclamide, glipizide and tolbutamide) in healthy volunteers. Fluconazole and oral sulphonylureas may be co-administered to diabetic patients, but the possibility of a hypoglycaemic episode should be borne in mind.
Phenytoin
Concomitant administration of fluconazole and phenytoin may increase the levels of phenytoin to a clinically significant degree. If it is necessary to administer both drugs concomitantly, phenytoin levels should be monitored and the phenytoin dose adjusted to maintain therapeutic levels.
Oral contraceptives
Two kinetic studies with combined oral contraceptives have been performed using multiple doses of fluconazole. There were no relevant effects on either hormone level in the 50mg fluconazole study, while at 200mg daily the AUCs of ethinyloestradiol and levonorgestrel were increased 40% and 24% respectively. Thus multiple dose use of fluconazole at these doses is unlikely to have an effect on the efficacy of the combined oral contraceptive. In a 300mg once weekly fluconazole study, the AUCs of ethinyl estradiol and norethindrone were increased by 24% and 13%, respectively.

Endogenous steroid
Fluconazole 50mg daily does not affect endogenous steroid levels in females: 200-400mg daily has no clinically significant effect on endogenous steroid levels or on ACTH stimulated response in healthy male volunteers.
Cyclosporin
A kinetic study in renal transplant patients found fluconazole 200mg daily to slowly increase cyclosporin concentrations. However, in another multiple dose study with 100mg daily, fluconazole did not affect cyclosporin levels in patients with bone marrow transplants. Cyclosporin plasma concentration monitoring in patients receiving fluconazole is recommended.
Theophylline
In a placebo controlled interaction study, the administration of fluconazole 200mg for 14 days resulted in an 18% decrease in the mean plasma clearance of theophylline. Patients who are receiving high doses of theophylline or who are otherwise at increased risk for theophylline toxicity should be observed for signs of theophylline toxicity while receiving fluconazole, and the therapy modified appropriately if signs of toxicity develop.
Terfenadine
Because of the occurrence of serious dysrhythmias secondary to prolongation of the QTc interval in patients receiving other azole antifungals in conjuntion with terfenadine, interactions studies have been performed. One study at a 200mg daily dose of fluconazole failed to demonstrate a prolongation in QTc interval. Another study at a 400mg and 800mg daily dose of fluconazole demonstrated that fluconazole taken in multiple doses of 400mg per day or greater significantly increased plasma levels of terfenadine when taken concomitantly. There have been spontaneously reported cases of palpitations, tachycardia, dizziness and chest pain in patients taking concomitant fluconazole and terfenadine where the relationship of the reported adverse events to drug therapy or underlying medical conditions was not clear. Because of the potential seriousness of such an interaction, it is recommended that terfenadine not be taken in combination with fluconazole.
Cisapride
There have been reports of cardiac events including torsades de pointes in patients to whom fluconazole and cisapride were co-administered. A controlled study found that concomitant fluconazole 200mg once daily and cisapride 20mg four times a day yielded a significant increase in cisapride plasma levels and prolongation of QTc interval. In most of these cases, the patients appear to have been predisposed to arrihythmias or had serious underlying illnesses, and the relationship of the reported events to a possible fluconazole –cisapride drug interaction is unclear. Because of the potential seriousness of such an interaction, co-administration of cisapride is contra-indicated in patients receiving fluconazole.
Zidovudine
Two kinetic studies resulted in increased levels of zidovudine most likely caused by the decreased conversion of zidovudine to its major metabolite. One study determined zidovudine levels in AIDS or ARC patients before and following fluconazole 200mg daily for 15 days. There was a significant increase in zidovudine AUC (20%). A second randomised, two-period, two-treatment coss-over study examined zidovudine levels in HIV infected patients. On two occasions, 21 days apart, patients received zidovudine 200mg every eight hours either with or without fluconazole 400mg daily for seven days. The AUC of zidovudine significantly increased (74%) during co-administration with fluconazole. Patients receiving this combination should be monitored for the development of zidovudine-related adverse reactions.
Rifabutin
There have been reports that an interaction exists when fluconazole is administered concomitantly with rifabutin, leading to increased serum levels of rifabutin. There have been reports of uveitis in patients to whom fluconazole and rifabutin were co-administered. Patients receiving rifabutin and fluconazole concomitantly should be carefully monitored.
Tacrolimus
There have been reports that an interaction exists when fluconazole is administered concomitantly with tacrolimus, leading to increased serum levels of tacrolimus. There have been reports of nephrotoxicity in patients to whom fluconazole and tacrolimus were co-administered. Patients receiving tacrolimus and fluconazole concomitantly should be carefully monitored. The use of fluconazole in patients concurrently taking astemizole or other drugs metabolised by the cytochrome P450 system may be associated with elevations in serum levels of these drugs. In the absence of definitive information, caution should be used when co-administering fluconazole. This is particularly important for drugs known to prolong QT interval. Patients should be carefully monitored. Interaction studies have shown that when oral fluconazole is co-administered with food, cimetidine, antacids or following total body irradiation for bone marrow transplantation, no clinically significant impairment of fluconazole absorption occurs. Physicians should be aware that drug-drug interaction studies with other medications have not been conducted, but that such interactions may occur.

Pregnancy and Lactation
Use during pregnancy
There have been reports of spontaneous abortion and congenital abnormalities in infants whose mothers were treated with 150mg of fluconazole as a single or repeated dose in the first trimester. Use in pregnancy should be avoided except in patients with severe or potentially life-threatening fungal infections in whom FLUCONOL® may be used if the anticipated benefit outweighs the possible risk to the fetus. If this drug is used during pregnancy, or if the patient becomes pregnant while taking the drug, the patient should be informed of the potential hazard to the fetus. Effective contraceptive measures should be considered in women of child-bearing potential and should continue throughout the treatment period and for approximately 1 week (5 to 6 half-lives) after the final dose. There have been reports of multiple congenital abnormalities in infants whose mothers were treated with high-dose (400mg/day to 800mg/day) fluconazole therapy for coccidioidomycosis (an unapproved indication). The relationship between fluconazole use and these events is unclear. Adverse fetal effects have been seen in animals only at high-dose levels associated with maternal toxicity. There were no fetal effects at 5 mg/kg or 10 mg/kg; increases in fetal anatomical variants (supernumerary ribs, renal pelvis dilation) and delays in ossification were observed at 25 mg/kg and 50 mg/kg and higher doses. At doses ranging from 80 mg/kg (approximately 20-60 times the recommended human dose) to 320 mg/kg, embryolethality in rats were increased and fetal abnormalities included wavy ribs, cleft palate and abnormal craniofacial ossification. Case reports describe a distinctive and a rare pattern of birth defects among infants whose mothers received high dose (400-800mg/day) fluconazole during most or all of the first trimester of pregnancy. The features seen in these infants include brachycephaly, abnormal facies, abnormal calvarial development, cleft palate, femoral bowing, thin ribs and long bones, arthrogryposis, and congenital heart disease.
Use during lactation
Fluconazole is found in human breast milk at concentrations similar to plasma. Breast-feeding may be maintained after a single dose of 150mg fluconazole. Breast-feeding is not recommended after repeated use or after high-dose fluconazole.

Side Effects
Fluconazole is generally well tolerated. The most common undesirable effects observed during clinical trials and associated with fluconazole are :
Nervous System Disorders : Headache
Skin and Subcutaneous Tissue Disorders : Rash
Gastrointestinal Disorders : Abnormal pain, diarrhoea, flatulence, nausea.
In some patients, particularly those with serious underlying diseases such as AIDS and cancer, changes in renal and haemotological function test results and hepatic abnormalities have been observed during treatment with fluconazole and comparative agents, but the clinical significance and relationship to treatment is uncertain.
Hepatobiliary Disorders : Hepatic toxicity including rare cases of fatalities, elevated alkaline phosphatase, elevated bilirubin, elevated SGOT, elevated SGPT.
In addition, the following undesirable effects have occurred during post-marketing :
Nervous System Disorders : Dizziness, seizures, taste perversion.
Skin and Subcutaneous Tissue Disorders : Alopecia, exfoliative skin disorders including Stevens-Johnson syndrome and toxic epidermal necrolysis.
Gastrointestinal Disorders : Dyspepsia, vomiting.
Blood and Lymphatic System Disorders : Leukopenia including neutropenia and agranulocytosis, thrombocytopenia.
Immune System Disorders :
Allegic reaction : Anaphylaxis (including angioedema, face oedema, pruritus), urticaria.
Hepatobiliary Disorders : Hepatic failure, hepatitis, hepatocellular necrosis, jaundice.
Metabolism and Nutrition Disorders : Hypercholesterolaemia, hypertriglyceridaemia, hypokalaemia.
Cardiac Disorders : QT prolongation, torsade de pointes.

Symptoms And Treatment of Overdose
There have been reports of overdosage with fluconazole and in one case, a 42 year-old patient infected with human immunodeficiency virus developed hallucinations and exhibited paranoid behaviour after reportedly ingesting 8200mg of fluconazole, unverified by his physician. The patient was admitted to the hospital and his condition resolved within 48 hours. In the event of overdosage, supportive measures and symptomatic treatment, with gastric lavage if necessary, may be adequate. As fluconazole is largely excreted in the urine, forced volume diuresis would probably increase the elimination rate. A three hour haemodialysis session decreases plasma levels by approximately 50%.

Storage Condition
Do not store above 30°C. Do not freeze.

Shelf Life
2 years from manufacturing date in the proper storage condition. Do not use after expiry.

Presentation
50mL X 20 LDPE plastic bottle per Carton.
100mL X 20 LDPE plastic bottle per Carton.

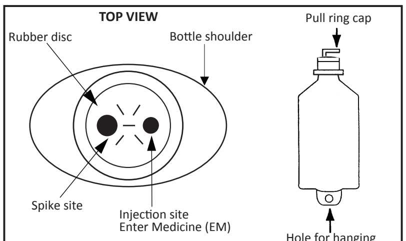
Manufacturer / Product Registration Holder:
AIN MEDICARE SDN. BHD.
Lot 4933, 4934 & PT2464, Jalan 6/44, Kaw. Perindustrian Pengkalan Chepa 2, 16100 Kota Bharu, Kelantan, MALAYSIA.

**** Keep out of reach of children/jauhkan daripada kanak-kanak***

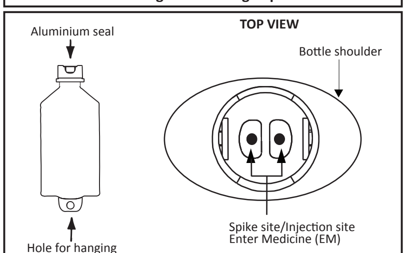
HANDLING INSTRUCTION

AIN MEDICARE INTRAVENOUS INFUSION FLUID IN LOW DENSITY POLYETHYLENE (LDPE) BOTTLE
The bottle is made from injectable grade polyethylene.

PARTS OF THE PLASTIC BOTTLE

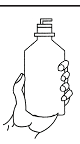


Design 1: Pull Ring Cap

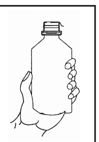


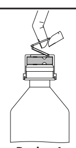
Design 2: Twin Port Cap (Aluminum Seal)

Top view after pull ring cap is removed. Please refer whichever applicable design.
INSTRUCTION FOR THE USAGE OF IV ADMINISTRATION SET & ADDITION OF DRUG


Design 1

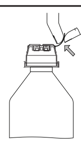
1. Before use check for any solid particle, growth, turbidity or leakage. Discard the product if particle, growth, turbidity or leakage is found.

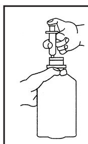

Design 2


Design 1

2. Design 1: Pull the cap's ring until completely removed.

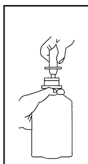
   Design 2: Pull the aluminum seal until completely removed by centre line.


Design 2

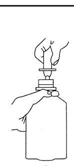

Design 1

3. Additives may be injected through injection site (Refer to Design 1 or 2 whichever applicable) on the rubber. Mix the solution thoroughly.


Design 2


Design 1

4. Hold bottle down firmly and insert a spike of administration set into the spike site for Design 1 OR for Design 2, pull the other aluminium seal until completely removed before inserting a spike of administration set.


Design 2

(Spike site is meant for spiking of the administration set and is only for single use)

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